

510K: A BENCHMARK FOR LATENT RELATIONSHIP INFERENCE AND WHEN-TO-COOPERATE IN MULTI-AGENT RL

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ABSTRACT

Multi-agent benchmarks typically hand the agent its social structure: teams are fixed and known, and only the question of *how* to cooperate remains. Realistic settings instead require deciding *whether and with whom* to cooperate, by inferring relationships from observed behaviour. We present **510K**, a four-player shedding card game built to make this latent-relationship inference problem measurable while remaining coupled to immediate scoring, terminal competition, and partial observability. Team identity and team size (2v2 vs 1v3) are determined at deal time by a private red-A distribution and hidden from observations. 510K offers four independently ablatable modes (SINGLE, STATIC, DYNAMIC, OBVIOUS), a graded information-reveal dial, Gymnasium and PettingZoo interfaces with action masking, and reference baselines. We formalize the resulting challenge as a *belief-inference gap*: the relationship is decodable from public play (probe AUC up to 1.00; logically determined for 36–39% of decisions), yet policies learn to defer only to a fixed, known partner (deferral asymmetry +0.38, $p < 10^{-4}$) and do not condition on a hidden partner (+0.00, $p = 0.21$) or an explicitly revealed one (−0.04, $p < 10^{-4}$), even though the reveal dial is a near-lossless information channel ($\geq 91\%$ of ideal mutual information). Structural probes show that the axes are not separable: scoring and finishing trade off, and the hidden team leaves most reward variance unattributable. A 1M-step, 5-seed masked-PPO suite, a recurrent variant, and a MAPPO baseline corroborate the failure. 510K is offered as a benchmark for *when-to-cooperate* decisions under coupled objectives.

1 INTRODUCTION

A central question in multi-agent reinforcement learning (MARL) is how agents should behave when several objectives interact and part of the task structure is hidden. Card games are a natural home for this question, yet existing benchmarks tend to isolate the difficulties one at a time. Dou Dizhu (Zha et al., 2021) couples competition with *fixed* cooperation and a sparse terminal reward. Hanabi (Bard et al., 2020) is purely cooperative, with neither competition nor an immediate scoring trade-off. Mahjong (Li et al., 2020) has scoring and partial observability but no teams. Tichu, Hearts and Spades have scoring and partnerships, but the partnerships are *fixed and known*. The picture that emerges is fragmented: we can measure how an algorithm handles scoring, or hidden information, or cooperation, but not how it handles them *together*.

This gap matters because in realistic tasks these aspects are not independent: acting on one changes the value of the others. A coding agent that makes a test pass now may damage the architecture later; an agent interacting with others must simultaneously infer who is aligned with it and whether that alignment persists. Benchmarks built by stacking independent switches cannot expose such interactions, because their optimal values decompose.

We introduce **510K**, a four-player shedding card game designed so that they do not. 510K couples (i) *immediate scoring* (5/10/K cards collected in won tricks), (ii) a *terminal competitive* objective (finish first), (iii) *partial observability*, and (iv) *dynamic cooperation*: team membership is determined at deal time by the private red-A distribution and is hidden, so “who should I help?” becomes a latent-variable inference problem rather than a given. Crucially, the axes are not stacked but

entangled — a single play changes the immediate score, the future pattern space, the information revealed about relationships, and the race to finish. We make this precise by formalizing coupling as *non-separability* and by giving each coupling a measurable signature rather than a rhetorical one (Section 6.3).

The environment ships with four controllable modes (SINGLE, STATIC, DYNAMIC, OBVIOUS) that ablate the cooperation and information axes, a graded information-reveal dial, Gymnasium and PettingZoo AEC interfaces with built-in action masking, and a reference baseline suite. On it we report a capability profile and a behavioural result: policies learn to cooperate with a fixed, known partner but neither infer a hidden partner nor exploit an explicitly revealed one — an information-utilization failure that mirrors the previously reported IIGC phenomenon (IIGC, 2027).

Contributions.

1. **A complete, tested environment.** Gymnasium single-agent and PettingZoo AEC interfaces with built-in action masking, four controllable modes, a deterministic rule bot, and 82 passing tests.
2. **A challenge decomposition and controlled variants** in which scoring, terminal competition, partial observability, and dynamic cooperation are independently ablatable.
3. **A coupling characterization.** We formalize “coupled, not stacked” as non-separability and give measurable structural signatures for the objective–objective, objective–cooperation, and cooperation–observability couplings.
4. **A baseline suite and evaluation protocol.** Masked PPO, recurrent masked PPO, and independent (shared-policy) PPO, evaluated identically as player 0 against fixed bots, with masking compliance as a correctness check and 95% confidence intervals plus significance tests for all comparisons.
5. **A capability profile and a cooperation-inference result.** At 1M steps and 5 seeds SINGLE is hardest for every method, team-mode win rates are inflated by teammate and team-selection effects, and a behavioural probe shows policies cooperate with a *fixed, known* partner but neither infer nor exploit a *hidden/revealed* one.

2 RELATED WORK

2.1 CARD-GAME RL TOOLKITS

RLCard (Zha et al., 2020) provides many card games behind a common interface and is the most widely used toolkit in this space; each included game, however, isolates a subset of challenges, and none exposes a hidden or dynamically determined team assignment. OpenSpiel (Lanctot et al., 2019) offers game-theoretic tooling over 50+ games, oriented toward solving extensive-form games rather than RL under sparse rewards. PyTAG (Balla et al., 2023) covers 20+ tabletop games with a modern API but shallow per-game depth. These toolkits make card games easy to run; they do not provide a controlled instrument for studying *interactions* between challenges.

2.2 SHEDDING-TYPE AND SCORING CARD GAMES

DouZero (Zha et al., 2021) established that a shedding-type game with a fixed landlord role and sparse terminal reward is a serious benchmark: “long horizons and sparse reward . . . the only time a nonzero reward is incurred is at the end of a game.” 510K keeps that long-horizon, sparse structure and adds two ingredients: *hidden, deal-time-determined teams* and an *immediate-scoring* objective that competes with the terminal finish objective. Tichu, Hearts and Spades have scoring and partnerships, but those partnerships are fixed and known, so the relationship itself is never inferred.

2.3 COOPERATIVE AND IMPERFECT-INFORMATION BENCHMARKS

The Hanabi Challenge (Bard et al., 2020) is the canonical cooperative imperfect-information benchmark — agents cannot see their own cards — but it has no competition and no scoring trade-off.

Table 1: Positioning: ● strong/present, ⊙ partial, ○ absent.

Challenge	510K	DouDizhu	Hanabi	Mahjong	Tichu	Overcooked	SMAC
Immediate scoring	●	○	○	●	●	○	○
Terminal competition	●	●	○	●	●	○	●
Partial observability	●	●	●	●	⊙	○	⊙
Team cooperation	● _{hidden}	● _{fixed}	● _{all}	○	● _{fixed}	●	⊙
Hidden/dynamic teams	●	○	○	○	○	○	○
Controlled ablation	●	○	○	○	○	⊙	⊙

Overcooked-AI (Carroll et al., 2019) probes ad-hoc human–AI coordination under full observability. Melting Pot (Agapiou et al., 2022) studies mixed-motive social interaction at a general substrate level, not within a compact, fully specified decision problem. None of these supplies a small, ground-truth, trainable instance in which the relationship between agents must be inferred.

2.4 MULTI-OBJECTIVE AND HIDDEN-ROLE SETTINGS

Multi-objective RL and planning methods tackle trade-offs between conflicting objectives (Hayes et al., 2022); 510K is a card-game instance with exactly two such objectives (immediate score and terminal finish) whose tension is measurable. Diplomacy/Cicero (Meta FAIR Diplomacy Team et al., 2022) and hidden-role LLM studies (Avalon, Werewolf) demonstrate strong interest in relationship inference at a much larger scale, but they lack a lightweight RL testbed with ground-truth team assignments. 510K fills that gap: the latent relationship is known to the simulator, so “did the agent infer it?” is directly measurable.

2.5 PRIOR RESULTS ON 510K

IIGC (IIGC, 2027) studies a *phenomenon* on this environment: under hidden team assignment, policy-gradient updates from different relationships cancel in expectation, producing *deceptive stability* (the policy appears to stop learning without converging). The present paper is deliberately complementary: IIGC asks *why* a learner stalls; we characterize the *environment* and provide the controlled ablations and behavioural probes that make the phenomenon reproducible and the mechanism studyable.

2.6 POSITIONING

Table 1 compares 510K with the environments above. Only 510K couples immediate scoring, terminal competition, partial observability, *hidden dynamic teams*, and controllable ablation in one game. The claim is about this combination and its controllability, not about being the hardest card game: each individual challenge is studied elsewhere, and our contribution is to make them jointly controllable and measurable.

3 THE 510K ENVIRONMENT

3.1 RULES

510K is a shedding-type card game played with a standard 52-card deck (4 players; a 3-player variant with two jokers is also implemented). Each player is dealt 13 cards. On a turn a player must either play a combination that beats the current trick or pass; when all other active players pass, the trick ends and its accumulated *score* is awarded to the trick leader, who opens the next trick. The result is a deliberately mixed reward structure: a dense, immediately observable score signal layered on a sparse terminal objective.

Immediate scoring. Cards score when collected in a won trick: 5 = 5 points, 10 = 10 points, K = 10 points. Each trick thus carries an immediate-score signal that competes with the game-level objective of emptying one’s hand first.

Terminal objective. The game ends when (SINGLE) the first player empties their hand, or (team modes) when both members of a team finish. Final reward combines collected 510K points with finish-position bonuses.

Patterns. 16 playable combination types: singles, pairs, triples, three-with-kicker, three-with-pair, straights, consecutive pairs, airplanes (with/without kickers), bombs, the eponymous **510K** (5-10-K, suited and non-suited), joker bomb, and (in team modes) red-A singles/pairs.

3.2 TEAM MECHANICS: HIDDEN DYNAMIC COOPERATION

- **STATIC:** fixed teams $\{0, 2\}$ vs $\{1, 3\}$; identities known.
- **DYNAMIC:** teams are determined at deal time by the red-A distribution: any player holding a red ace (or all four aces) joins the “red” team. Team membership is *hidden* from observations and must be inferred from others’ play.
- **OBVIOUS:** same deal-time teams as DYNAMIC, but the observation includes a 4-bit team mask — a direct ablation of teammate-identity visibility.

The hidden relationship can be a 2v2 partnership or a 1v3/3v1 solo configuration: the red team has size 2 with probability $1 - 4 \binom{50}{11} / \binom{52}{13} \approx 0.765$ and size 1 otherwise — a closed-form hypergeometric fact verified exactly (Section 6.9). We treat the solo case as a feature: the latent variable to infer is not only *who* is my partner but *whether I even have one*. Because the team is hidden, the reward function itself is latent: the same observation–action pair can map to different returns depending on the deal.

3.3 INTERFACES

- **Gymnasium single-agent** (`FiveTenKEnv`, registered as `510K-v0`; Towers et al., 2024): controls player 0; other players are auto-played by a random or rule bot. Observation is a normalized 112-dim vector (own-hand one-hot + last-trick cards/type + hand sizes + current player + pass count + own score; +4 team bits in OBVIOUS). 300-discrete action space with built-in masking; passes `gymnasium.utils.env_checker` in all four modes.
- **PettingZoo AEC** (`FiveTenKMultiEnv`; Terry et al., 2021): controls all players; observation is an `{observation, action_mask}` dict per agent; passes the official `api.test`. We provide AEC (not Parallel) because 510K is strictly turn-based with irregular turn order.
- **Action masking** is built in; the largest hand produces ≈ 75 valid patterns, far below the 300 action slots.

3.4 REPRODUCIBILITY

Seeded `reset`, deterministic rules, `pip install env-510k`, 82 passing tests, and a published reference implementation (`baselines/`). A trajectory release and Docker configuration are planned.

4 CHALLENGE DECOMPOSITION AND CONTROLLED VARIANTS

The central design claim: 510K does not *stack* challenges; they **interact and alter each other’s value**. Playing a 5/10/K card changes the immediate score, the remaining hand and future pattern space, may reveal or hide information about team membership, and affects who can finish first. This yields a difficulty hierarchy: immediate reward $\rightarrow$ situation value $\rightarrow$ teammate/opponent intent $\rightarrow$ future state space $\rightarrow$ game outcome.

4.1 COUPLING IS NON-SEPARABILITY, NOT STACKING

A benchmark *stacks* challenges when a difficulty (or the optimal value) decomposes additively across axes. Let the axes be scoring (S), terminal competition (T), partial observability (P) and hidden cooperation (C). The couplings we deliberately instantiate are:

Table 2: Controlled variants (the experimental backbone).

Axis	SINGLE	STATIC	DYNAMIC	OBVIOUS	Reveal
Scoring	✓	✓	✓	✓	✓
Terminal competition	✓	✓	✓	✓	✓
Teams	none	fixed, known	hidden	hidden, revealed	DYNAMIC + frac.
Partial observability	✓	✓	✓	✓	graded

- **C1 objective entanglement ($S \times T$).** A 5/10/K play moves the immediate score and the race to finish *in tension*: securing points means keeping control of the trick, while shedding fast concedes points.
- **C2 cooperation $\times$ observability ($C \times P$).** In DYNAMIC the reward is indexed by the latent team T ; two histories that look identical can induce different reward mappings. Partial observability is *constitutive of* the cooperation structure, not additive noise.
- **C3 objective $\times$ cooperation ($S \times C$).** The same scoring objective has a different payoff depending on team structure.
- **C4 information $\times$ cooperation $\times$ state.** The value of revealing the team depends on whether the agent has a partner (1v3 vs 2v2) and on the terminal race; `RevealEnv` turns this into a graded dial.

Because every axis is a code-level switch, each coupling has a *measurable* signature (Section 6.3) rather than a rhetorical one.

Controlled variants. The information-reveal axis is realized by `RevealEnv`, which includes the team mask with probability p per decision ($p=0 \equiv$ DYNAMIC, $p=1 \equiv$ OBVIOUS).

5 EXPERIMENTAL SETUP

Baselines (all action-masked). **Masked PPO (MLP):** `sb3_contrib.MaskablePPO` (Raffin et al., 2021) (PPO; Schulman et al., 2017) controlling player 0 against fixed auto-played opponents (random or rule bot). **Recurrent PPO (LSTM):** `RecurrentPPO` (Raffin et al., 2021) + a custom `MaskedLstmActorCriticPolicy` (`sb3-contrib` ships no masked LSTM policy; we embed the mask in the observation and apply it inside the policy); its evaluation collapses in DYNAMIC/OBVIOUS (Section 6.4 and Section 7). **Independent PPO (IPPO):** a shared-policy self-play baseline (`SelfPlayEnv`): one policy controls all seats.

Evaluation protocol. Every policy is evaluated identically: the trained policy plays player 0 against fixed opponents for seeded games. Metrics: win rate, mean total reward (mean $\pm$ 95% CI), mean episode length, and illegal-action rate (masking compliance).

Protocol. All runs are 1M steps in modes SINGLE/STATIC/DYNAMIC/OBVIOUS: masked PPO with 5 seeds, IPPO with 3, recurrent PPO with 2. Capability-profile analysis asks which method collapses under (hidden team $\times$ partial observability).

Coupling and cooperation-inference probes. Structural coupling signatures are measured with fixed bots driving all four seats (`coupling_probes.py`): a score-finish rank correlation (C1) and a reward-attribution decomposition (C2/C3). Strategy-level behaviour is probed with trained policies (`analyze_team_conditioning.py`) via a deferral asymmetry (Section 6.4).

Statistical analysis. Cells are reported as mean $\pm$ 95% CI (t-based, `ddof=1`). We use Welch t -tests for algorithm comparisons, paired t -tests for within-checkpoint comparisons, a per-seed linear slope of win rate on reveal probability p with a one-sample t -test, and two-proportion z -tests for the deferral asymmetries (`baselines/stats.py`).

Compute budget. Measured on a single GPU: `MaskablePPO` $\approx$ 500 env-steps/s ($\approx$ 40 min per 1M-step run); the 4-mode $\times$ 5-seed masked-PPO matrix costs $\approx$ 13 GPU-hours.

Table 3: Capability profile. [†]Recurrent checkpoints remain seed-degenerate in DYNAMIC/OBVIOUS.

mode	MLP PPO (5 seeds)	recurrent PPO (2) [†]	IPPO (3)
SINGLE	0.516 $\pm$ 0.029	0.540	0.300 $\pm$ 0.114
STATIC	0.774 $\pm$ 0.031	0.735	0.737 $\pm$ 0.094
DYNAMIC	0.862 $\pm$ 0.016	0.850 [†]	0.810 $\pm$ 0.025
OBVIOUS	0.822 $\pm$ 0.006	0.780 [†]	0.807 $\pm$ 0.080

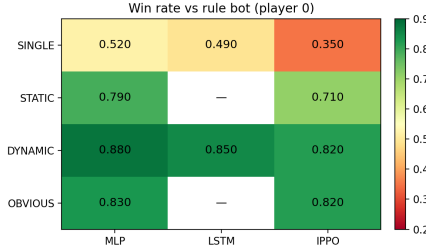


Figure 1: Capability profile (win rate vs rule bot, player 0).

6 RESULTS

Core results use masked PPO trained for 1M steps with 5 seeds and evaluated as player 0 against rule bots. All policies have illegal-action rate 0.0 — masking is respected exactly.

6.1 CAPABILITY PROFILE ACROSS MODES AND METHODS

Win rate (fraction of episodes with positive team reward), mean $\pm$ 95% CI (all 1M steps). Recurrent is the `ent_coef=0.05` variant (2 seeds, no CI).

Reading: **SINGLE is the hardest setting** for every method (MLP 0.516 $\pm$ 0.029, IPPO 0.300 $\pm$ 0.114, below chance). **MLP $\hat{!}$ IPPO in all four modes**, significantly in SINGLE (Welch $p = 0.007$) and DYNAMIC ($p = 0.001$) but not in STATIC ($p = 0.22$) or OBVIOUS ($p = 0.50$). Team-mode win rates are inflated by two effects: a teammate’s play contributes to the outcome, and DYNAMIC/OBVIOUS teams are *selected* by holding a red ace, so the agent’s team tends to hold stronger cards; we therefore treat per-agent and behavioural metrics as the primary evidence. DYNAMIC is significantly above OBVIOUS (0.862 vs 0.822, paired $t = 8.94$, $p = 0.0009$), but the difference is small and opponent-dependent (it reverses against random bots).

Opponent robustness. Re-evaluating the same rule-trained checkpoints against random bots (`evaluate_opponents.py`) preserves the ordering: SINGLE stays hardest (MLP 0.764, IPPO 0.450) and MLP $\hat{!}$ IPPO in every mode (STATIC 0.950/0.867, DYNAMIC 0.948/0.907, OBVIOUS 0.958/0.950; full table in Appendix A). Team modes saturate near ceiling against random bots, so the rule bot is the more informative opponent.

6.2 INFORMATION-REVELATION ABLATION

Masked PPO with the team revealed with probability p per decision (`RevealEnv`), all at 1M steps $\times$ 5 seeds:

Against rule-bot opponents the curve is flat to slightly decreasing. A per-seed linear fit gives a mean slope of -0.030 per unit p (one-sample $t = -6.12$, $p = 0.0036$, $n = 5$), and the paired $p=0$ vs $p=1$ difference is $+0.040$ ($p = 0.0009$): at 1M steps, revealing the team significantly *hurts* rather than helps. Prop. C shows the dial is a nearly-linear information channel ($\geq 91\%$ of the ideal $pH(T)$), so this is a genuine *information-utilization* failure, not an unturned dial. Against random bots the difference reverses and is within noise, so the opponent-robust claim is the behavioural one.

Table 4: Information-reveal dose-response (MLP).

reveal p	win rate	reward
0.00 (DYNAMIC)	0.862 ± 0.015	78.0 ± 5.0
0.25	0.838 ± 0.020	79.3 ± 4.3
0.50	0.838 ± 0.019	76.9 ± 3.4
0.75	0.844 ± 0.029	77.7 ± 3.4
1.00 (OBVIOUS)	0.822 ± 0.005	69.6 ± 0.1

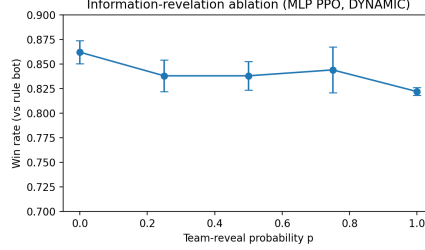


Figure 2: Information-revelation dose-response (MLP, DYNAMIC).

6.3 STRUCTURAL COUPLING SIGNATURES (C1–C3)

Fixed-bot probes over 500 games/mode. top1st = fraction of games whose top scorer finishes first (chance 0.25, binomial test); ρ = within-game rank correlation between score and finish position (mean $\pm$ 95% CI, one-sample t -test); $dR/d(\text{own})$ = OLS slope of reward on own score; R^2 = reward variance explained by own+teammate scores (bootstrap 95% CI).

The random bot shows the same pattern (e.g. SINGLE $R^2 = 0.638$ [0.61, 0.66] vs DYNAMIC 0.200 [0.17, 0.23]). C1 holds: scoring and finishing first genuinely trade off. C3 holds: $\text{corr}(\text{own score}, \text{reward})$ drops from ≈ 0.8 in SINGLE to 0.44–0.59 in team modes. C2 holds: in hidden-team modes only 0.20–0.36 of reward variance is attributable to own+teammate scores, versus 0.64–0.84 in SINGLE/STATIC (non-overlapping bootstrap CIs for SINGLE vs DYNAMIC).

6.4 CAN A POLICY CONDITION ON THE HIDDEN TEAM? (C2/C4)

On every following decision we measure the deferral asymmetry $\text{asym} = P(\text{pass} \mid \text{true partner leads}) - P(\text{pass} \mid \text{opponent leads})$.

A fixed, known partner is learned; a hidden partner is not inferred (null); and merely revealing the partner does not help — OBVIOUS is significantly *negative* and its win rate is lower than DYNAMIC. Four of the five OBVIOUS seeds produce identical decision sequences on the shared evaluation deals, a signature of a policy that has collapsed to ignoring the team bits.

Recurrent and CTDE corroboration. The MLP is memoryless, so we also trained two masked-LSTM variants and a MAPPO baseline with a centralized critic and self-play (1M steps). The healthiest LSTM is seed-diverse in SINGLE (0.540 ± 0.030) and STATIC and learns fixed-partner cooperation ($+0.37/+0.38$), yet shows no cooperation under hidden or revealed teams (DYNAMIC -0.043 ; OBVIOUS $-0.07/-0.06$). MAPPO likewise shows no positive deference to the hidden partner (DYNAMIC $\text{asym} -0.021$, range -0.025 to -0.016 , 3 seeds) despite much higher win rates (0.93–0.98) — so the failure is neither memoryless nor PPO-specific, and team win rate is again a poor metric.

6.5 THE BELIEF-INFERENCING GAP

We formalize the challenge as a *belief-inference gap*: the latent relationship z (who, if anyone, is a teammate) must be inferred from the public history h_t , and the policy should condition on it. A linear probe over the agent’s features predicts the true teammate set with AUC 0.60/0.91

Table 5: Structural couplings, rule bot (all $\rho < 0$ and all top1st > 0.25 at $p < 10^{-9}$).

mode	top1st	ρ (95% CI)	$dR/d(\text{own})$	corr(own, R)	R^2 (95% CI)
SINGLE	0.596	-0.271 ± 0.043	1.51	0.893	0.797 [0.78, 0.82]
STATIC	0.592	-0.546 ± 0.038	2.34	0.587	0.844 [0.82, 0.87]
DYNAMIC	0.604	-0.565 ± 0.036	2.16	0.569	0.337 [0.30, 0.37]
OBVIOUS	0.374	-0.364 ± 0.046	2.23	0.588	0.356 [0.32, 0.39]

Table 6: Deferral asymmetry; p -values are one-sample t -tests against zero.

mode	MLP (5 seeds)	recurrent e05 (2)
STATIC	$+0.379$ ($p < 10^{-4}$)	$+0.375$
DYNAMIC	$+0.003$ ($p = 0.21$)	-0.043
OBVIOUS	-0.037 ($p < 10^{-4}$)	-0.066

(DYNAMIC/OBVIOUS) from a single observation, rising to 0.70/1.00 with history, and the relation is *logically determined* by public information for 39%/36% of decisions; yet the trained policy’s deferral asymmetry is $+0.003$ (DYNAMIC, $p = 0.21$) and -0.037 (OBVIOUS, $p < 10^{-4}$). The relationship is decodable from public play but unused.

6.6 HIDDEN-RELATIONSHIP ROBUSTNESS (1v3 VS 2v2)

DYNAMIC deals split by red-team size (76% 2v2 / 24% 1v3). Over the five seeds the 2v2 and 1v3 win rates are 0.840 (range 0.82–0.85) and 0.891 (0.85–0.93); the deal split is 384/116 $\approx$ 76.8%/23.2%, matching Prop. A. The results are insensitive to team composition, so conclusions do not depend on the intentionally unfiltered 1v3 deals.

6.7 LEARNING CURVES

Per-chunk evaluation history (Figure 3) shows that every mode plateaus early: the mean slope over the last three chunks is not different from zero in any mode ($|\text{slope}| \leq 0.004$ per 100k steps, all $p > 0.24$, $n = 5$ seeds). SINGLE never leaves chance (final 0.516 ± 0.029 vs 0.5, $p = 0.195$); DYNAMIC and OBVIOUS are statistically flat by 300k already (final 0.862/0.822 vs 0.836/0.796, paired $p = 0.13$ and 0.07). The information-utilization failure is therefore not a “not yet trained” artifact.

6.8 CONNECTION TO IIGC

Under hidden teams, PPO exhibits *deceptive stability* (IIGC, 2027): at 1M steps its evaluation performance is unchanged whether or not the team is revealed, and the behavioural probe shows it never conditions on the true partner — while the OBVIOUS ablation does not help and slightly hurts. This benchmark paper supplies the environment characterization behind that phenomenon.

6.9 THEORETICAL GROUNDING

Four analytical results tie the environment’s design and the observed results together:

1. **Team-size prior (design).** The red team has size 2 with probability $1 - 4 \binom{50}{11} / \binom{52}{13} \approx 0.765$ and size 1 otherwise — a derived hypergeometric fact (verified exactly over 20k deals). The latent relationship includes both identity and size.
2. **Membership probabilities.** $P(\text{agent} \in \text{red}) \approx 0.441$, and $P(\text{agent is the solo red player}) \approx 0.059$.
3. **The reveal dial is a nearly-linear information channel:** $MI(p) = p H(T) - \delta(p)$, with $\delta(p) = 0$ iff no deal is 1v3; measured $MI = 0.62/1.26/1.91$ at $p = 0.25/0.5/0.75$ ($\geq 91\%$ of ideal), so the axis *was* swept.

Table 7: Belief-inference gap (1200 games/mode, 13.9k/17.8k agent decisions).

mode	determinable	obs-probe AUC	hist-probe AUC	policy asym
DYNAMIC	0.39	0.60	0.70	+0.003
OBVIOUS	0.36	0.91	1.00	−0.037

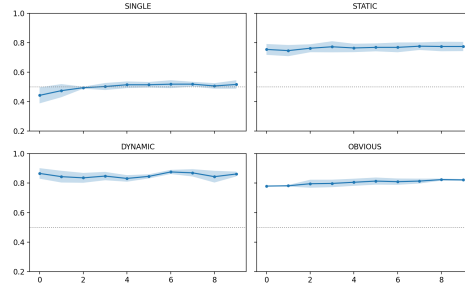


Figure 3: Learning curves per mode (MLP, mean $\pm$ 95% CI over 5 seeds).

4. **Reward dilution explains coarse win rates:** team-mode rewards aggregate two players’ scores, and the teammate contributes $\approx$ half the variance (own share $\approx$ 0.51), so team win rate is a threshold on a diluted signal.

7 DISCUSSION, LIMITATIONS, AND FUTURE WORK

510K is not claimed to be the “hardest card game”; it is a *controllable combination* of individually-studied challenges — a unified testbed rather than a SOTA chase. Unlike social benchmarks with opaque relationship structure, it has ground-truth deal-time teams, so “did the agent infer the relationship?” is directly measurable; our deferral asymmetry realizes this and finds that current PPO does *not* condition on the true partner.

Limitations.

- *Coverage.* The masked-PPO matrix and the full reveal curve are complete at $1M \times 5$ seeds and IPPO at $1M \times 3$ seeds; the recurrent baselines cover all modes at 2 seeds but remain partly degenerate, and opponents are rule/random bots (no self-play or search).
- *Coarse win rate and recurrent collapse.* Team modes share the win with teammates and select stronger teams, so per-agent metrics are the primary signals; masked-LSTM seeds collapse to a seed- and state-independent “play-lowest-valid” policy (after fixing an eval-path bug), and higher-entropy variants rescued seed diversity in SINGLE/STATIC but not DYNAMIC/OBVIOUS.
- *A negative result.* Whether a stronger opponent or a different algorithm eventually exploits the revealed information is open.

Future work. More seeds, reward-shaping stability (the 510K score as a dense reward competing with the terminal objective (Ng et al., 1999; Skalse et al., 2022)), MAPPO/QMIX, search/planning baselines, human and LLM-agent play, a 3-player mode, and a trajectory dataset.

8 CONCLUSION

510K couples immediate scoring, terminal competition, partial observability, and — uniquely — *hidden dynamic cooperation* in one controllable card game. We formalize the coupling as non-separability and give measurable signatures: objectives trade off, the hidden team injects non-attributable variance, and a 1M-step policy cooperates with a known partner but neither infers nor exploits a hidden or revealed one — echoing IIGC’s deceptive stability.

REPRODUCIBILITY STATEMENT

- **Environment:** `pip install env-510k (Gymnasium + PettingZoo AEC, gymnasium.register('510K-v0'))`; 82 tests.
- **Baselines & analysis:** `baselines/` with fixed seeds, chunked checkpoints, and per-chunk eval history.
- **Compute:** one GPU; ≈ 500 env-steps/s for MaskablePPO (≈ 40 min per 1M-step run; the $4\text{-mode} \times 5\text{-seed}$ matrix ≈ 13 GPU-hours).
- **Data:** `per-run *.eval.json` and `*.history.jsonl`, plus coupling / positions / team-conditioning / inferability analyses and the statistical report (`runs/stats-report.md`).

REFERENCES

- Balla, M., Long, G. E. M., Jeurissen, D., Goodman, J., Gaina, R. D., Perez-Liebana, D. PyTAG: Challenges and Opportunities for Reinforcement Learning in Tabletop Games. IEEE Conference on Games (CoG) 2023. arXiv:2307.09905.
- Bard, N., Foerster, J., Chandar, S., Burch, N., Lanctot, M., Song, H. F., Parisotto, E., Dumoulin, V., Moitra, S., Hughes, E., Dunning, I., Mourad, S., Larochelle, H., Bellemare, M. G., Bowling, M. The Hanabi Challenge: A New Frontier for AI Research. Artificial Intelligence 280, 2020.
- Carroll, M., Shah, R., Ho, M. K., Griffiths, T. L., Seshia, S. A., Abbeel, P., Dragan, A. On the Utility of Learning about Humans for Human-AI Coordination. NeurIPS 2019.
- Agapiou, C., Vezhnevets, A. S., Duéñez-Guzmán, E. A., Matyas, J., Mao, Y., Sunehag, P., Köster, R., Madhushani, U., Kummerfeld, E., et al. Melting Pot 2.0. arXiv:2211.13746, 2022.
- Hayes, C. F., Rădulescu, R., Bargiacchi, E., Källström, J., Macfarlane, M., Reymond, M., Verstraeten, T., Zintgraf, L. M., Dazeley, R., Heintz, F., et al. A Practical Guide to Multi-Objective Reinforcement Learning and Planning. Autonomous Agents and Multi-Agent Systems 36(26), 2022.
- Lanctot, M., Lockhart, E., Timbers, F., Davoodi, S., Lewis, M., Cheng, B., De Bono, V., Graepel, T., Bachrach, Y. OpenSpiel: A Framework for Reinforcement Learning in Games. arXiv:1908.09453, 2019.
- Li, J., Koyamada, S., Ye, Q., Liu, G., Wang, C., Yang, R., Zhao, L., Qin, T., Liu, T.-Y., Hon, H.-W. Suphx: Mastering Mahjong with Deep Reinforcement Learning. arXiv:2003.13590, 2020.
- Meta Fundamental AI Research Diplomacy Team (FAIR), Bakhtin, A., Brown, N., Dinan, E., et al. Human-level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning. Science 378(6624):1067–1074, 2022.
- Ng, A. Y., Harada, D., Russell, S. Policy Invariance Under Reward Transformations: Theory and Application to Reward Shaping. ICML 1999.
- Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., Dormann, N. Stable-Baselines3: Reliable Reinforcement Learning Implementations. Journal of Machine Learning Research 22(268):1–8, 2021.
- Zha, D., Lai, K.-H., Cao, Y., Huang, S., Wei, R., Guo, J., Hu, X. RLCard: A Toolkit for Reinforcement Learning in Card Games. AAAI-20 Workshop on Reinforcement Learning in Games, 2020. arXiv:1910.04376.
- Samvelyan, M., Rashid, T., de Witt, C. S., Farquhar, G., Nardelli, N., Rudner, T. G. J., Hung, C.-M., Torr, P. H. S., Foerster, J., Whiteson, S. The StarCraft Multi-Agent Challenge. AAMAS 2019.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.

Skalse, J., Howe, N. H. R., Krashennnikov, D., Krueger, D. Defining and Characterizing Reward Hacking. NeurIPS 2022.

Terry, J. K., Black, B., Grammel, N., Jayakumar, M., Hari, A., Sullivan, R., Santos, L. S., Diefendahl, C., Horsch, C., Perez-Vicente, R., Williams, N., Lokesh, Y., Ravi, P. PettingZoo: Gym for Multi-Agent Reinforcement Learning. NeurIPS 2021 (Datasets & Benchmarks).

Towers, M., Kwiatkowski, A., Terry, J., Balis, J. U., De Cola, G., Deleu, T., Goulão, M., Kallinteris, A., Krimmel, M., KG, A., et al. Gymnasium: A Standard Interface for Reinforcement Learning Environments. arXiv:2407.17032, 2024.

Zha, D., Xie, J., Ma, W., Zhang, S., Lian, X., Hu, X., Liu, J. DouZero: Mastering DouDizhu with Self-Play Deep Reinforcement Learning. ICML 2021.

Information-Induced Gradient Contraction in Partially-Observable MARL. Prior work, AAAI 2027.

A RANDOM-BOT EVALUATION

Table 8 reports the same rule-trained checkpoints evaluated against random bots. Team modes saturate near ceiling (0.87–0.96); the rule bot is therefore the more informative opponent and all main-text claims use it.

Table 8: Rule- vs random-bot win rate (same checkpoints, mean over seeds).

mode	MLP rule	MLP random	IPPO rule	IPPO random
SINGLE	0.516	0.764	0.300	0.450
STATIC	0.774	0.950	0.737	0.867
DYNAMIC	0.862	0.948	0.810	0.907
OBVIOUS	0.822	0.958	0.807	0.950

B 1v3 VS 2v2 ROBUSTNESS BY SEED

Table 9 gives the per-seed split summarised in Section 6.6.

Table 9: DYNAMIC win rate by red-team size (500 eval games/seed).

seed	2v2 win rate	1v3 win rate
0	0.820	0.897
1	0.852	0.931
2	0.854	0.853
3	0.846	0.871
4	0.828	0.905